

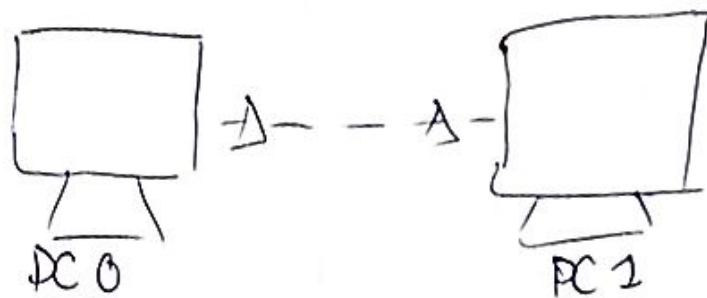
Practical-1

Aim: To study basic commands and perform basic network configuration using Cisco Packet Tracer.

Networking commands help in checking connection configuration and troubleshooting. Commands like ipconfig, ping and traceret are used to verify IP configuration.

Steps

- 1) Open Cisco Packet Tracer.
- 2) Drag 2 PCs (PSO and PCI) into workspace.
- 3) Connect them using copper cross-over cable.
- 4) Click on PC0 → Desktop → IP Configuration.
- 5) Assign IP Address: 10.20.30.1
- 6) Click on PC1 → Desktop → IP configuration.
- 7) Assign IP Address 10.20.30.2
- 8) Open command prompt in PC0
- 9) Type commands
ipconfig
ping
tracert



PC > ipconfig

IP address 10.20.30.1

Subnet mask 255.0.0.1

Default Gateway 0.0.0.0

PC > ping 10.20.30.2

Reply from 10.20.30.2 : bytes=32 time<1ms

Reply from 10.20.30.2 : bytes=32 time<1ms

Reply from 10.20.30.2 : bytes=32 time<1ms

Ping statistics: Sent=4, Received=4, Succeeded=4 (0% loss)

PC > tracert 10.20.30.2

Tracing route to 10.20.30.2 over a max of 3 hops

1 0 ms 0 ms 0 ms 10.20.30.2

Trace complete

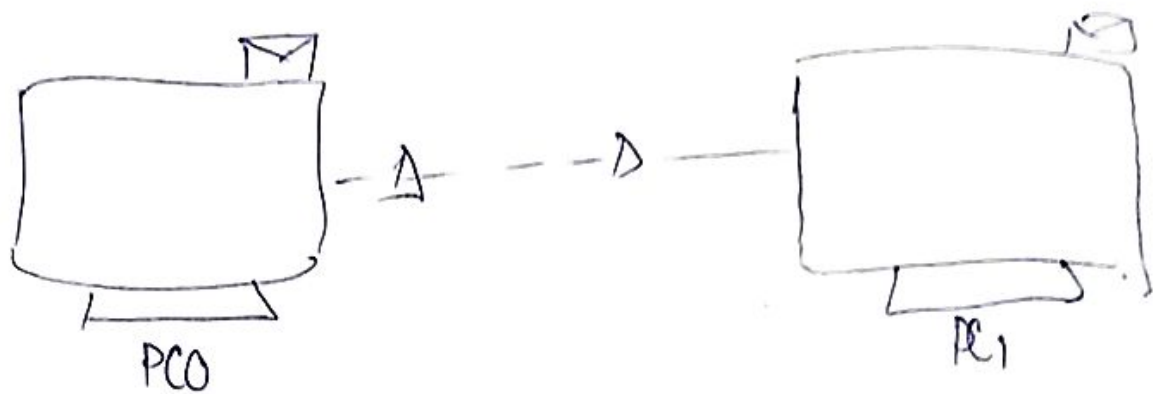
Practical-2

Aim: To study and perform PC to PC communication in Cisco Packet Tracer.

PC to PC communication allows data transfer between two systems using IP address and networking cables.

Steps

- Place 2 PCs (PC0, PC1)
- connect them using cross over cable
- Configure IP
 - PC0 : 10.20.30.1
 - PC1 : 10.20.30.2
- Test connectivity by ping
- Click on the envelope icon
- select source to destination
- Click on simulation and play.
- It will move from left to right.



PC > ping 10.20.30.2

Reply from 10.20.30.2 : bytes=32 time 1ms TTL=
" " " " "
" " " " "

Ping Statistics : sent = 4, Received = 4, lost = 0 (0% loss)

Practical-3

Aim: To design and stimulate a star topology using a switch and hub.

Steps

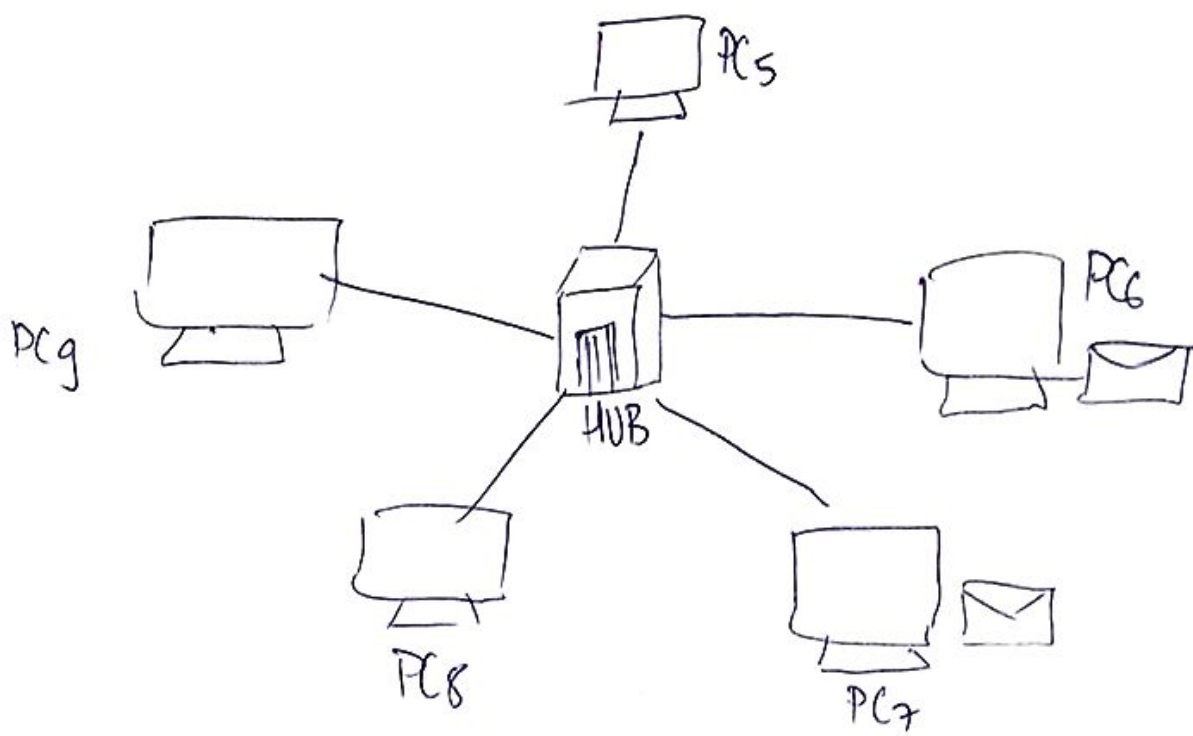
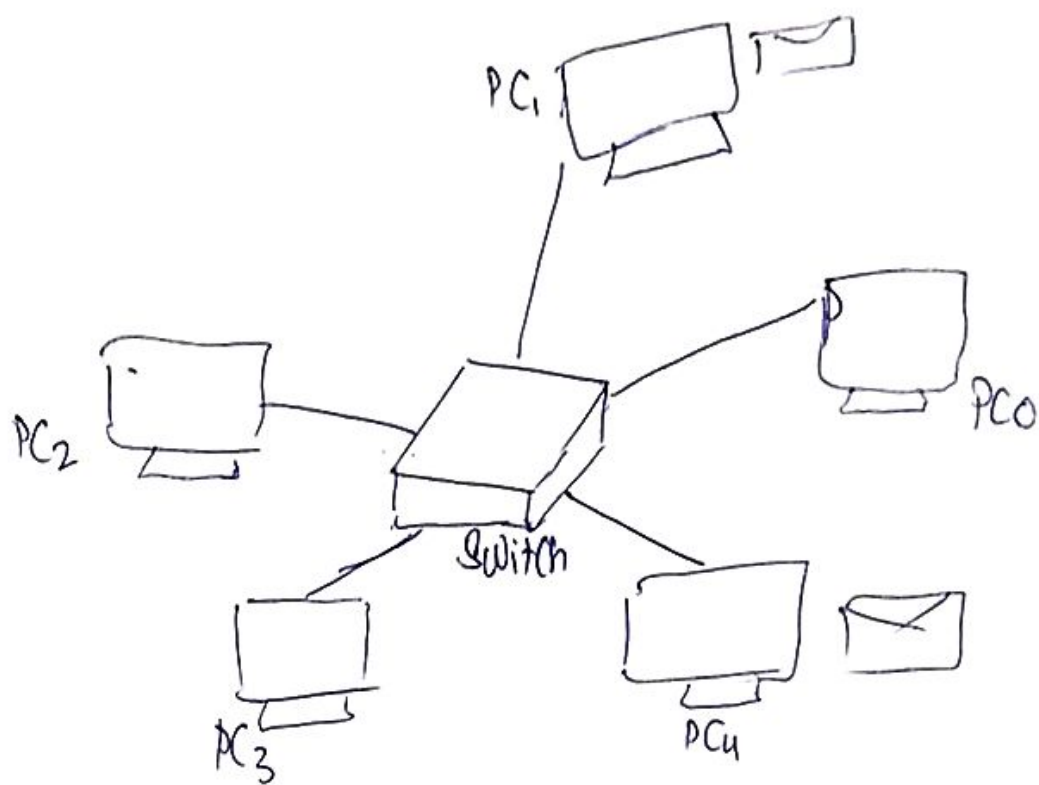
- Drag 5 PCs in the environment from end devices.
- Assign IP addresses.
- Place the PCs around the switch/hub.
- Click connection choose copper-straight-through wire.
- Connect all of them to HUB/switch.
- Click on envelope icon to send a message.
- Select Sender

Receiver

- Stimulate and Play

Conclusion

- Hub will broadcast the message to all but only receiver will accept the message.
- Switch will broadcast only the first time after that it will only send it to the receiver.



Practical-4

Aim: To Stimulate Bus, Ring, Mesh and Hybrid Topologies in Packet Tracer.

Bus Topology:

- End Devices → 4 of them → Assign IPs
- Network devices → 4 switches
- Place them linearly.
- Select copper ~~wireless~~ for switches - switch
- select copper - straight-through for switch-to-PC.
- Send message by envelope icon and stimulate.

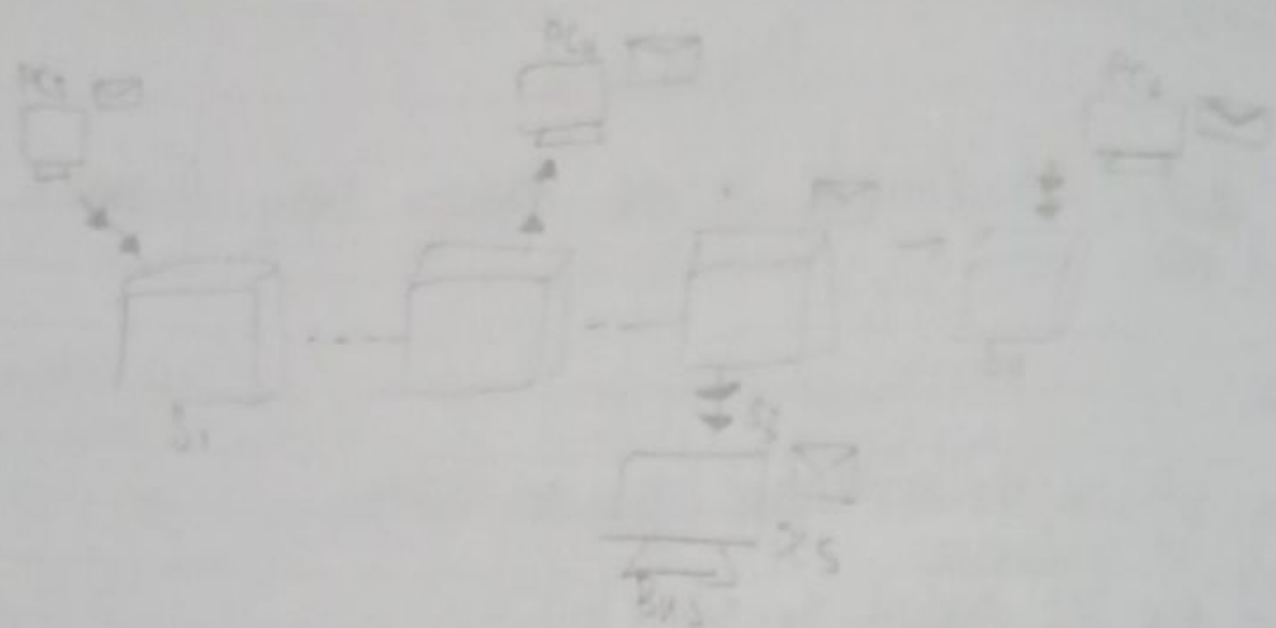
Observation: → Packet Travel linearly.
→ All nodes receive signal
→ Delay was higher.

Ring Topology:

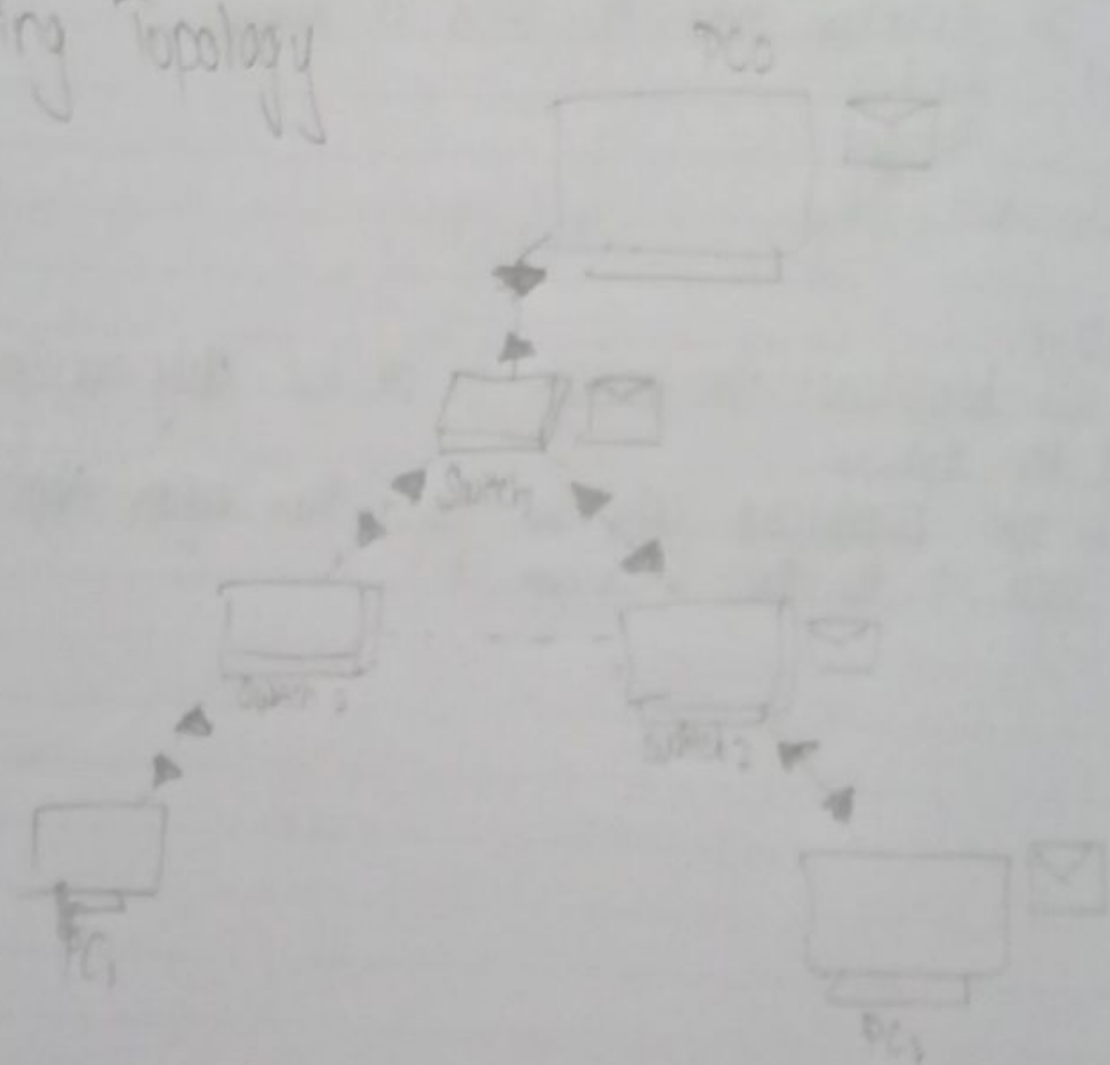
- End devices → PC 5 → 3 PC 3 → Assign IPs.
- Network device - switch - Place 3 to 2
- connect switches in Ring and PC to switches using copper cross over and straight-through wire respectively
- Send message and run stimulation.

Observation: → Each node forward the frame
→ No collision observed in simulation.

Bus Topology



Ring Topology



Hybrid Topology

- End Devices → PCs → Assign IP.
- Network Devices → Switches
- Instead of last PC extend its branch to fit bus topology linearly.
- Connect the cables → use copper cross for same type of device and copper straight for different
- Send message and run simulation

Observation:

- Combination of mesh and star.
- Broadcasters to star and moved linearly in bus.

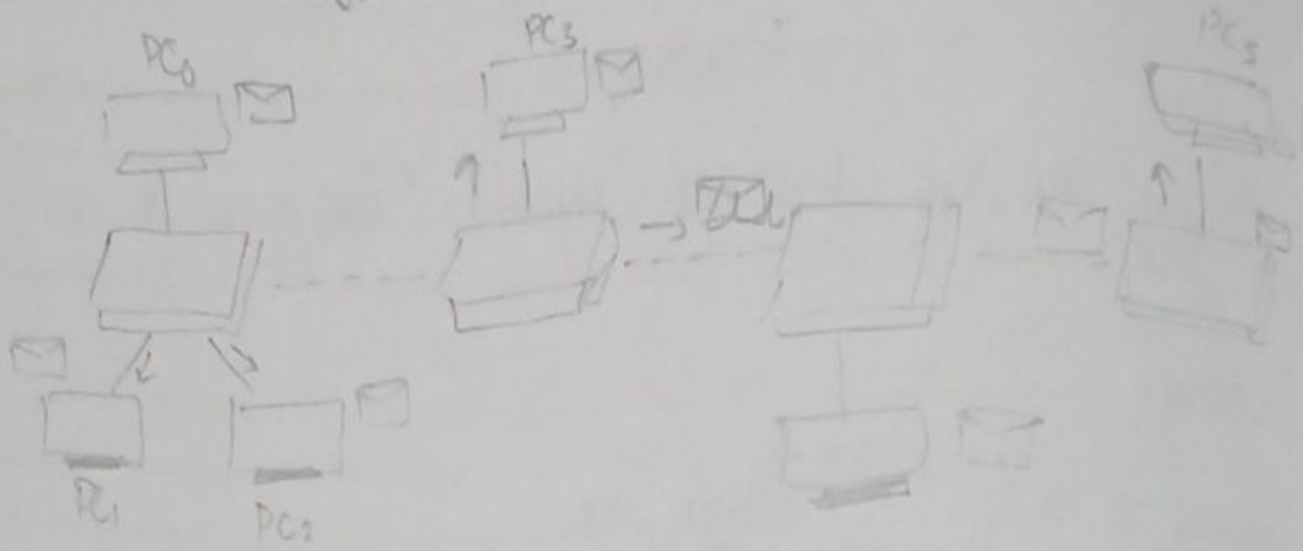
Mesh topology

- End Devices → PCs 5 → Assign IP
- Network device → Switch.
- Place switch around each other and place PCs around them.
- Connect switch to each other including diagonally using copper cross cable.
- Connect PCs to switches using copper straight wire.
- Send message & run simulation.

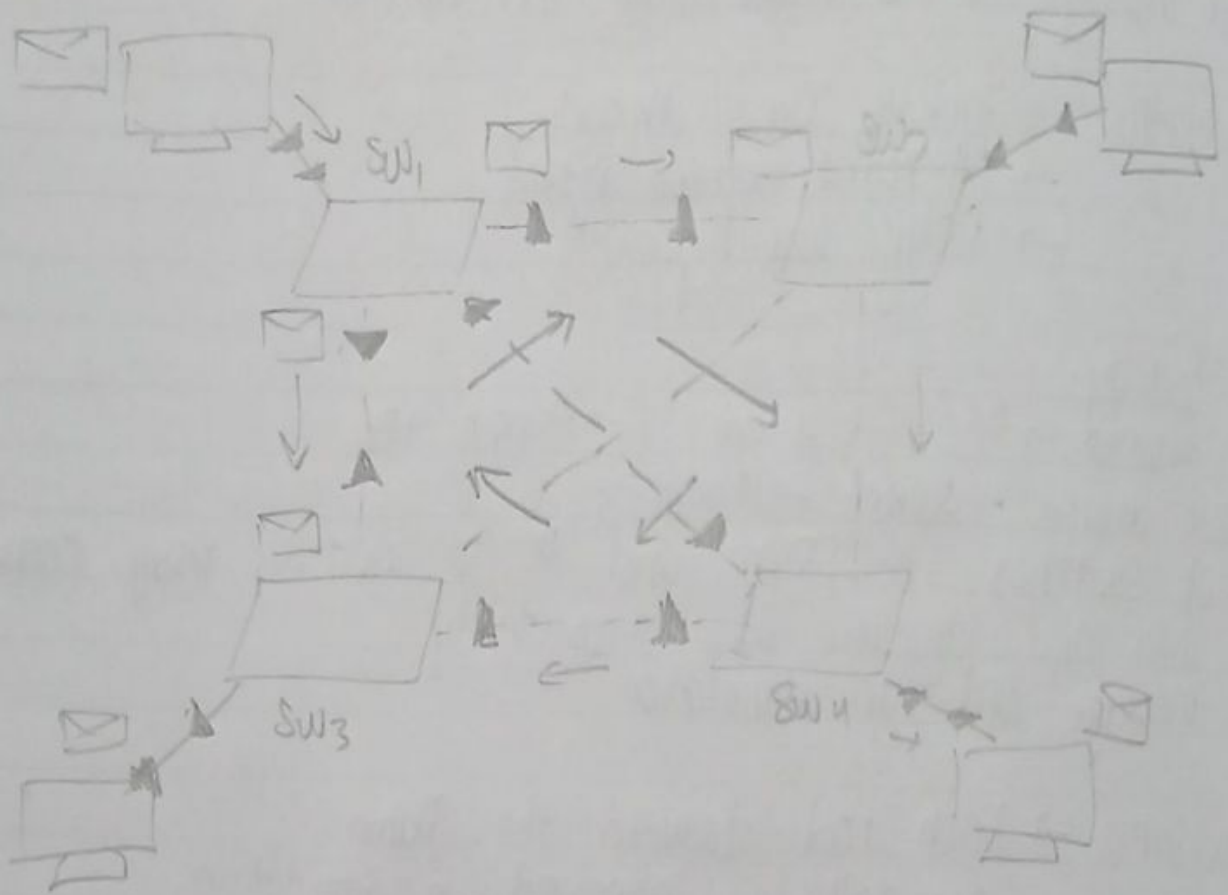
Observation: Multiple paths are available data. The packet chose shortest path.

Topology showed high reliability.

Hybrid



Mesh Topology



Practical-5

Aim: To perform initial switch configuration.

Steps: → Open packet tracer and place a switch.
→ Double click and go to "CLI" tab.

Configure hostname:

- 1.) Type enable in CLI to enable EXEC cable
- 2.) Type 'conf t' to change global configuration.
- 3.) Then type hostname [name] to change the name of switch to 'name'.

Configure message of the day (motd)

- 4.) Type banner motd \$
- 5.) Then type the message of the day.
- 6.) After you are done type '\$' at the end of message.

Setting up password

- 7.) Line console 0' to enable console line configuration.
- 8.) Password 123. Set the password to 123.
- 9.) Then type login and hit enter to enable login for line console.
- 10.) Type exit to leave line console.
- 11.) Type enable secret 123 to set up password for exec mode to '123'.
- 12.) Type exit to leave exec mode.

Changing hostname

```
Switch>enable  
Switch # conf t  
Switch (config) # hostname sw0  
sw0 (config) #
```

Configuring message of the day

```
sw0 (config) # banner motd $  
Enter TEXT message. End with character $  
!! only authorized Personnel Allowed !! $
```

Setting up password

```
sw0 (config) # line console 0  
sw0 (config-line) # password 123  
sw0 (config-line) # login  
sw0 (config-line) # exit  
sw0 (config) # enable secret 123  
sw0 (config) # exit  
sw0 #
```

After configuration

!! Only authorized personnel!

User Access Verification

7. Password

sw0>

Practical-6

Aim To perform an initial switch configuration.

Steps :

- Add a switch and a PC.
- Connect them using console cable.
- Open terminal in PC.
- It will prompt for initial configuration dialog.
- Enter 'yes'.
- Then enter the name of switch and press enter.
- Then you will be prompted to set up password for EXEC mode.
- Then you will be prompted to set up enable password.
- After that it will be prompted to set up enable password.
- After that it will be prompted to set up SNMP Network Management: type no and press enter. It will be done later manually when devices are connected.
- After that list of commands will be shown which will be executed.
- Type 2 to save the configurations.

Conclusion

The initial switch configuration has been performed. The hostname and password has been setup.



Would you like to enter initial configuration dialog? [yes/no]: yes

Would you like to enter basic management setup? [yes/no]:

yes

Enter hostname: q1

Enter enable secret: 123

Enter enable password: 123

Enter virtual terminal password: 12345

Configure SNMP Network Management? [no]: no

The following configuration script was generated:

!

host name q1

enable secret 5 \$ 1\$ 123\$

enable password 1234

...

[0] Go to IOS command prompt without saving this config.

[1] Return back to the setup without saving this config.

[2] Save this configuration to ram & exit.

Enter your selection [2]: 2

Practical - 7

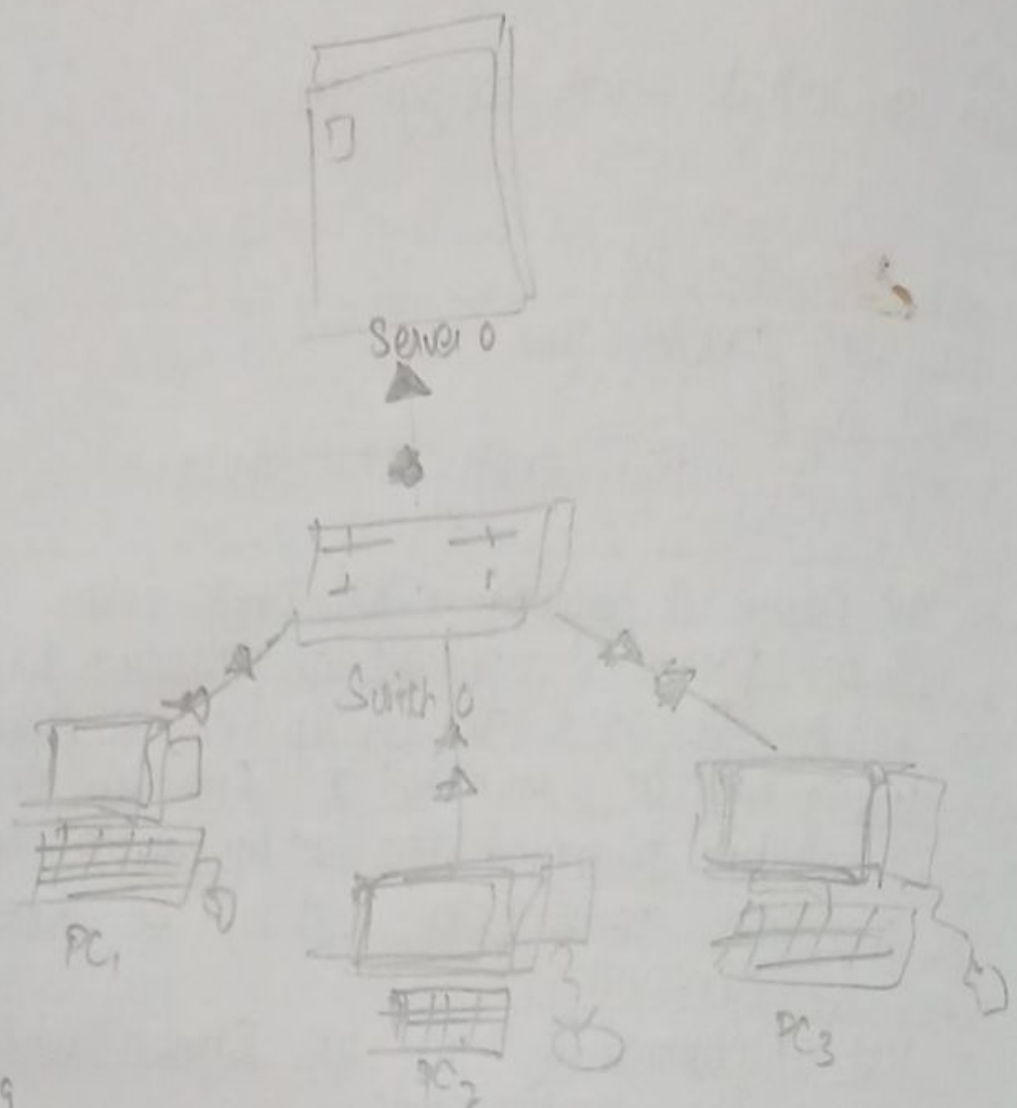
Aim: To implement client server network

Step

- Add a server from end devices.
- Add a switch from network devices.
- Add 3 PCs from end devices.
- connect them using copper wires.
- Double click server → config → Fast Ethernet 0
configure the IP address (192.168.0.1)
- Goto services → DHCP
Set up static IP address (192.168.0.1)
Set maximum no. of users : 512
Tick on to start the service.
- Go to IP configuration of PCs and select DHCP. IP address will be assigned automatically.
- To verify desktop → command prompt
Type ipconfig to see for ipaddress.

Observation

The end device got their IP addresses automatically from the server.



C : 1 ipconfig

Fast Ethernet 0 connection

IPv6 Address ::1

IPv4 Address 192.168.0.2

Subnet Mask 255.255.255.0

Practical-8

Aim: To implement connection between devices using a router.

Steps:

- To add a switch and connect three PCs to it.
- Add another switch and connect three PCs to it.
- Assign the IP addresses to PCs of first LAN (192.168.0.1, 192.168.0.2, 192.168.0.3)
- Assign IP address to PCs of second LAN (10.0.0.1, 10.0.0.2, 10.0.0.3)
- Add a router and connect the switches to it using Gigabit Ethernet.

→ Configure Router:

Config → Gigabit Ethernet 0/0/0:

add IPv4 Address 192.168.0.4

Set port status to on.

For second switch

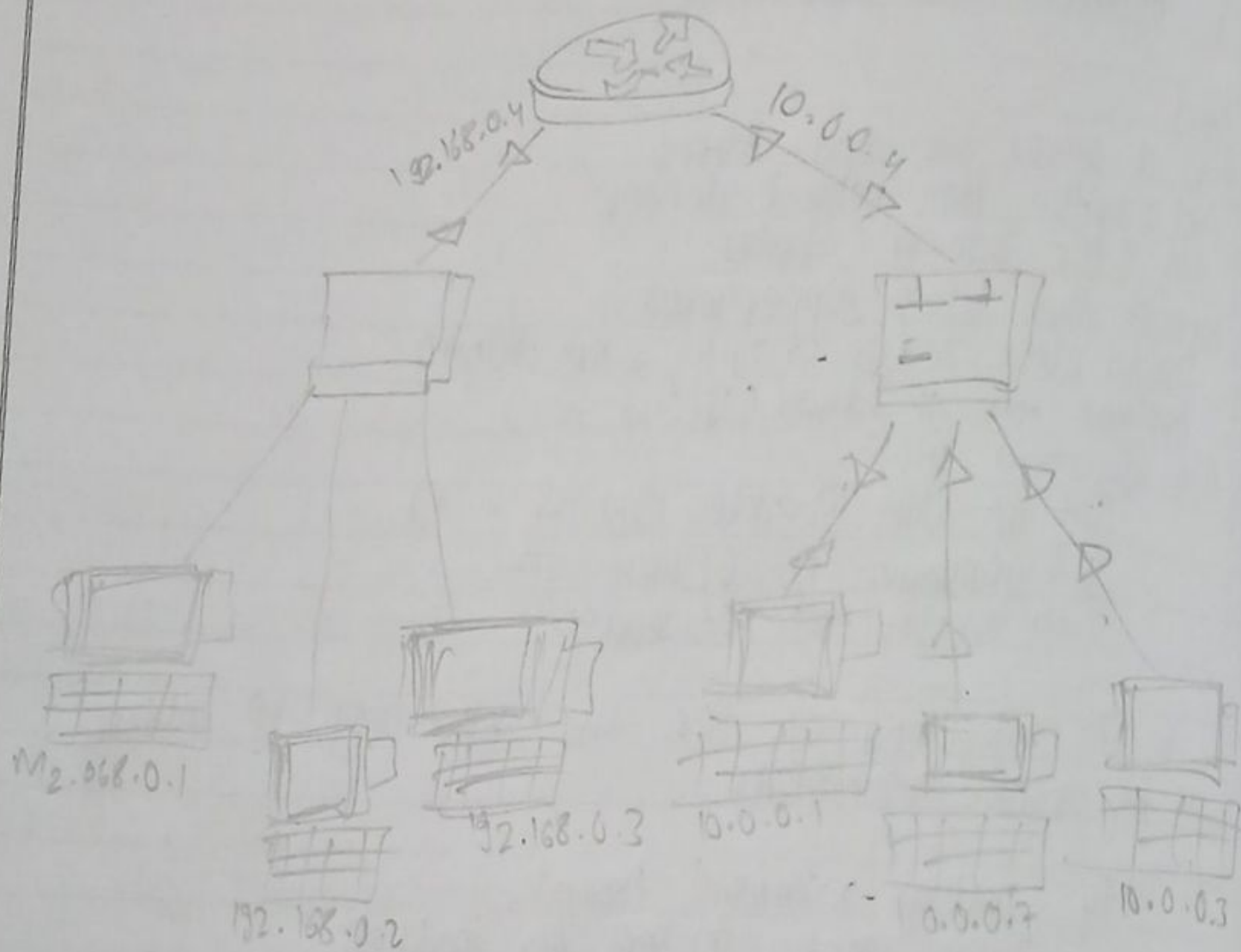
Config → Gigabit Ethernet 0/0/1

add IPv4 Address 10.0.0.4

Set port status to on.

- Set the default gateway of PCs in first LAN to 192.168.0.4. For PCs in second LAN set the default gateway to 10.0.0.4.

Observation:



Message can be successfully sent to the PCs of another LAN.

Conclusion:

Two LANs were successfully connected using a router.

C:\> ping 10.0.0.2

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.2: byte=32 time=1ms TTL=127
" " " " " "
" " " " " "

Ping statistics for 10.0.0.2:

Packets: Sent=4 Received=4, lost=0 (0% loss)

Practical - 9

Aim: To perform remote desktop within LAN connection.

Steps:

- Open Cisco Packet Tracer and drag and drop ~~two~~ two PCs and one Switch.
- Connect two PCs with the switch using copper straight-through wire.
- Configure IP address of PC0 (192.168.0.1) and PC1 (192.168.0.2).
- Open 'CLI' in the switch.
- Run the following commands:
enable
conf t
hostname cnamd
interface vlan 1
ip address 192.168.0.5 255.255.255.0
no shutdown
exit
username cisco password 123
line vty 0 4
login local
transport input telnet
- Now check the connection using ping.
- Open PC0 → command prompt.
- telnet 192.168.0.5

Switch CLI

```
Switch>enable
Switch#conf t
Switch(config)# hostname anshreeka
anshreeka(config)# interface vlan 1
anshreeka(config-if)# ip address 192.168.0.5 255.255.255.0
anshreeka(config-if)# no shutdown
anshreeka(config-if)# exit
anshreeka(config)# username cisco password 123
anshreeka(config)# login local
anshreeka(config-line)# transport input telnet
```

Command Prompt

```
C:\>telnet 192.168.0.5
```

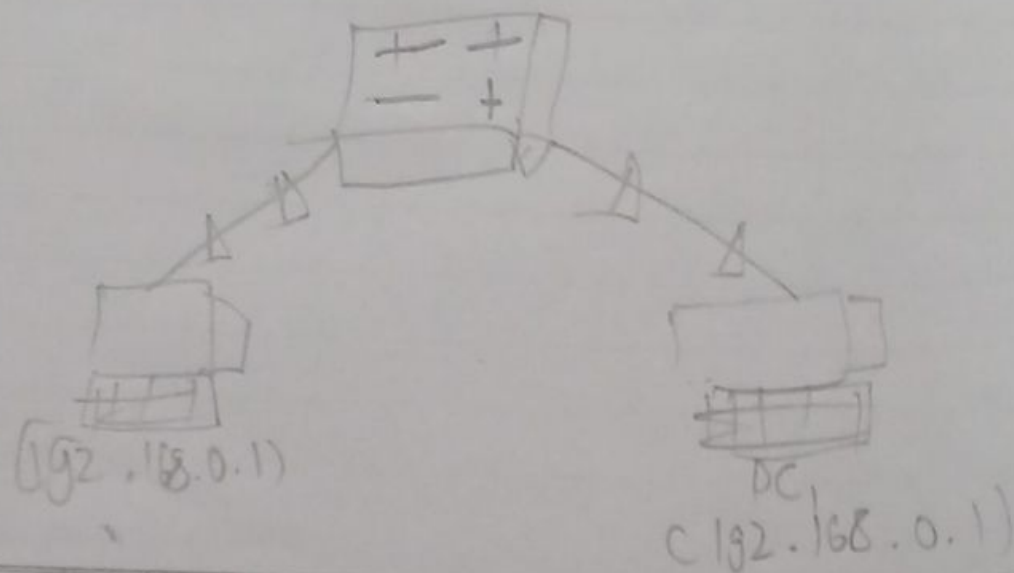
```
Trying 192.168.0.5 ... Open
```

User Access Verification

```
Username: cisco
```

```
Password:
```

```
anshreeka>
```



→ Now enter username and password set in the switch configuration.

→ Now we can remotely access the switch.

Conclusion:

We can remotely access the switch using the PC.